[YOUR NAME]

MECHANICAL ENGINEER

📍 [City, State] 📱 +91-XXXXXXXXXX ✉️ [your.email@example.com] 🔗 [linkedin.com/in/yourprofile]

PROFESSIONAL SUMMARY

Detail-oriented and results-driven **Mechanical Engineer** with a strong foundation in mechanical design, CAD modeling, thermal-fluid analysis, and manufacturing processes. Proficient in 3D CAD modeling (SolidWorks, AutoCAD), Finite Element Analysis (FEA), and Geometric Dimensioning and Tolerancing (GD&T). Adept at problem-solving, quality control, and cross-functional team collaboration to deliver efficient, safe, and cost-effective engineering solutions.

TECHNICAL SKILLS & CORE COMPETENCIES

Design & Software: SolidWorks, AutoCAD, CATIA, ANSYS (FEA/CFD), MATLAB, Fusion 360.

Manufacturing & Standards: GD&T (ASME Y14.5), CNC Machining, Injection Molding, Sheet Metal Design, Quality Assurance.

Engineering Concepts: Thermodynamics, Fluid Mechanics, Strength of Materials, Kinematics, Heat Transfer.

Methodologies & Soft Skills: Lean Manufacturing, Root Cause Analysis (RCA), 5S, Project Planning, Cross-Functional Teamwork.

WORK EXPERIENCE / INTERNSHIPS

Mechanical Engineering Intern

[Month Year] – [Month Year]

[Company / Industry Name] — [City, State]

- Assisted in designing mechanical components and assemblies using **SolidWorks**, ensuring compliance with manufacturing standards and **GD&T** tolerances.
- Conducted stress and thermal stress analysis using **ANSYS FEA**, optimizing component design to reduce material weight by 12% while maintaining structural integrity.
- Participated in root cause analysis (RCA) and assembly line troubleshooting, helping reduce production downtime by 10%.
- Prepared detailed technical documentation, Bill of Materials (BOM), and standard operating procedures (SOPs).

ACADEMIC & TECHNICAL PROJECTS

Design and Structural Optimization of an All-Terrain Vehicle (ATV) Frame

Major Capstone Project

Lead Design Engineer

- Modeled a 3D tubular space frame chassis in SolidWorks and performed impact analysis (Frontal, Rear, Side roll-over) using ANSYS.
- Selected optimal AISI 4130 chromoly steel tubing to achieve high strength-to-weight ratio and improved driver safety margins by 15%.

Automated Solar Tracking & Cooling Mechanism

Minor Engineering Project

Mechanical & Thermal Designer

- Designed a dual-axis mechanical tracking framework with integrated heat sink cooling, enhancing solar panel power efficiency by 18%.

EDUCATION & CERTIFICATIONS

B.Tech / B.E. in Mechanical Engineering

Graduation Year: [202X] | CGPA: [X.XX/10]

[University / College Name], [City, State]

Certified SolidWorks Associate (CSWA/CSWP) — Dassault Systèmes

ANSYS Workbench Certification — FEA & CFD Simulation